

Heart-Mind Prioritized and Narrative-Constrained Intelligent Recommendation Decision Framework

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Abstract. When introducing AI-based recommendation systems into holistic personality education, a fundamental contradiction emerges: the click-through rates and dwell time maximized by commercial recommendation logic often run counter to the heart-mind cultivation sought by education. Content that best captivates students frequently avoids real challenges to core qualities such as concentration, resilience, and discernment; while tasks genuinely capable of stretching students' heart-mind weaknesses may bring frustration and cognitive discomfort, scoring poorly in traditional recommendation models. To resolve this problem, this paper proposes a comprehensive intelligent recommendation decision framework. It first breaks through the conventional “cognition–interest” two-dimensional profile by constructing a four-dimensional student profile. On this basis, the framework reconstructs the utility function of the recommendation system, giving the decision weights of value activation and heart-mind targeting far greater importance than cognitive fit and interest match, thereby establishing a “growth first” value order at the algorithmic level. The core innovation of the Heart-Mind Bandit algorithm lies in replacing the traditional bandit’s immediate reward signal (clicks, dwell time) with cultivation gain quantified by physiological and behavioral data: resilience improvement is measured by the shortening of the half-life recovery time of heart rate variability after a setback; concentration enhancement is characterized by the rate of decrease in postural entropy and improvement in gaze stability; discernment growth is calibrated by the incidence of self-correction behaviors; and compassion development is captured through peer nomination frequency and increased inter-brain synchronization measured via hyperscanning. This work marks a profound transition of AI’s role in education from a “dessert chef” catering to desires to a “nutritionist” safeguarding growth, providing an interpretable and auditable paradigm for the responsible deployment of AI decision-making systems in the domain of values education.

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1. Introduction

Recommendation systems have become an indispensable decision-making core for digital platforms, continuously optimizing content ranking and distribution by predicting users' immediate behaviors such as clicks, views, and purchases [1]. However, when this set of technologies, guided by the creed of “maximizing engagement,” is transplanted into the context of red education and holistic personality cultivation, a profound paradox is exposed. Those contents that best capture students' attention and extend their dwell time—such as fast-paced historical animations or shallow quizzes requiring no deep thought—often precisely avoid real challenges to students' heart-mind weaknesses. In contrast, a resilience training task requiring a student to face repeated failures in a sandbox simulation alone, or a discernment review prompting a student to confront their own cognitive biases, may manifest as higher bounce rates and lower completion rates at the behavioral level, yet bear irreplaceable heart-mind cultivation value in the educational sense. If the reward signal of the recommendation algorithm remains anchored on clicks and dwell time, it will inevitably slide toward “entertainment-oriented indulgence,” becoming a “dessert chef” who merely serves palatable sweets, rather than a “nutritionist” prescribing for students' long-term growth.

The holistic personality education intelligent engine addressed in this paper has the fundamental mission not of allowing students to consume content within their comfort zone, but of promoting substantial development in the nine core heart-mind qualities — concentration, discernment, resilience, adaptability, reverence, critical thinking, compassion, perseverance, and creativity—through precise task recommendation. This requires a thorough reconstruction of the recommendation system's underlying decision logic: the optimization objective must be transformed from “maximizing behavioral participation” to “maximizing Nine-Force cultivation gain.” This reconstruction means that the algorithm must answer three progressively deepening questions. First, what is the student's current state of value identification and heart-mind development? Second, among numerous available instructional tasks, which one can most precisely challenge the qualities that are still weak? Third, how can a narrative path be planned over a teaching cycle spanning weeks or even months that maximizes cumulative heart-mind benefits while never violating the intrinsic logic of historical events?

In response to these questions, this paper makes four interrelated contributions. First, it proposes a four-dimensional student profile model of “value identification – heart-mind development – cognitive state – interest preference,” formally incorporating values and heart-mind qualities into the input space of the recommendation system, and designs a construction method based on multimodal perception fusion and dynamic confidence labeling. Second, it undertakes a heart-mind-oriented transformation of the contextual bandit algorithm, reconstructing the utility function and redefining the reward signal to quantify the Nine-Force cultivation gain produced by each recommendation round using physiological and behavioral data. Third, it introduces Monte Carlo Tree Search into knowledge graph path planning and formalizes three types of hard constraints — temporal, causal, and spatial — erecting insurmountable bottom lines for narrative seriousness within the search algorithm. Fourth, a prototype system was deployed in an actual red education practice base, and the framework’s dual advantages in heart-mind improvement and historical narrative consistency were validated through a combination of offline simulation and classroom trials, with discussion on the mechanism for preserving student autonomy within such AI-assisted decision-making.

2. Four-Dimensional Student Profiling

Traditional educational recommendation systems habitually describe learners as a two-dimensional vector composed of cognitive state and interest preference. The cognitive state vector includes indicators such as cognitive load, attentional margin, and knowledge mastery, answering “can the student learn it”; the interest preference vector captures the intensity of the student’s preference for different themes and forms, answering “is the student willing to learn” [1]. This framework can achieve better-than-random content matching in a general sense, but when facing red education with heart-mind cultivation as the core goal, it exposes a structural deficiency: it completely suspends the two decisive dimensions of students’ value identification and heart-mind qualities. A student may exhibit extremely high interest in a certain military history topic and possess reading ability sufficient to easily handle the complexity of related texts, yet may not have been truly activated in value dimensions such as responsibility or devotion to the people, and her resilience may be too weak to withstand any form of failure. If the recommendation system makes decisions based solely on interest and cognitive fit, it will continuously push content that is “liked and easy,” which is precisely the most insidious trap that keeps students lingering within their heart-mind comfort zone.

Therefore, this section expands the student profile from two dimensions to four, adding the value identification vector and the heart-mind development level vector, and repositioning the original cognition and interest vectors as auxiliary adjustment dimensions subordinate to value and heart-mind. These four dimensions are not a parallel list of features but follow a strict priority order: value identification anchors the direction, heart-mind development quantifies the foundation, cognitive state demarcates the boundary, and interest preference selects the entrance.

The data foundation of the value identification vector comes from the Red Nine-Soul subscale of the Holistic Personality Omnibus Scale, which measures the depth of students' identification with nine dimensions including loyalty, responsibility, devotion to the people, and pragmatism. To overcome the problem that traditional self-report scales are susceptible to social desirability bias, items in the scale are designed with behavioral anchors, requiring students to make choices in specific situations—such as conflicts between collective and personal interests or between actively assuming responsibility and avoiding it—rather than directly asking “Do you identify with loyalty?” Furthermore, we introduce two implicit enhancement mechanisms to continuously calibrate the value vector. The first is a behavioral consistency check: when a student listens to a revolutionary martyr's last letter, if her skin conductance signal shows significantly elevated cross-individual coupling strength with the narrator's audio signal, this physiological empathy indicator forms cross-validation with her high score on the “devotion to the people” subscale; conversely, if the system repeatedly detects a behavioral pattern of actively avoiding responsibility in team tasks, a cautious “behavioral inconsistency” mark is attached to the responsibility dimension, prompting the recommendation system to postpone pushing tasks with high responsibility requirements. The second mechanism is knowledge-graph-based association enhancement: when a student repeatedly retrieves and deeply reads historical figures highly associated with a certain Red Soul quality and their complete transmission paths during autonomous browsing, this exploratory behavior is interpreted as an early awakening signal for that quality, and the system accordingly applies a moderate positive fine-tuning to the corresponding dimension of the value vector.

The heart-mind development level vector is a quantified expression of the current state of the student's Nine Forces, constructed not from a single source of self-report but through weighted fusion of three-channel data. The first channel is the Nine-Force self-report scale score, providing the student's subjective cognition of their own heart-mind qualities. The

second channel consists of behavioral performance records embedded in OMOEC courses, such as sustained attention duration and interruption frequency during LEGO building tasks, the time to recover engagement and the frequency of strategy adjustments after setbacks in the Long March sandbox simulation, and the frequency of active helping behaviors and peer nomination frequency in team collaboration. The third channel comes from real-time heart-mind perception data output by the multimodal physiological computing module, including concentration marked by postural entropy and eye movement gaze stability [2], resilience characterized by the recovery rate of RMSSD and HF power of heart rate variability before and after setbacks [3], and compassion reflected by inter-brain synchronization collected through hyperscanning technology [4]. The fusion of three-channel data is not a simple equal-weight averaging but employs differentiated weighting based on the differential validity of each channel on different Nine-Force dimensions. For instance, on the concentration dimension, objective physiological indicators such as eye movement gaze stability and postural entropy carry relatively high weight; on the compassion dimension, peer nomination and actual helping behaviors are the core validity criteria, while the self-report scale weight is moderately reduced due to high social desirability bias. After fusion, each Nine-Force dimension is accompanied by a confidence label indicating the reliability of the current assessment. When the three-channel data are highly consistent, the confidence is labeled “high,” and the system can make decisive recommendations; when an inconsistent pattern of “high self-report, low behavior, low physiology” appears, the label is “to be confirmed,” and the system subsequently initiates heart-mind exploration; if a student has long avoided a certain type of task resulting in completely blank data for that force, the label is “data blank”—precisely the signal that the recommendation system needs to prioritize.

The cognitive state vector inherits and extends validated quantitative methods such as real-time cognitive load estimation and Bayesian knowledge tracing [5]; its core function is to set a difficulty ceiling for recommended tasks, ensuring that heart-mind challenges do not exceed the student’s current cognitive capacity. The interest preference vector continues to be modeled through content tag weights and collaborative filtering [6], capturing students’ acceptance tendencies for different narrative entrances such as military equipment, heroic figures, and daily life scenes. In the repositioned four-dimensional architecture, cognitive state and interest preference are explicitly placed below value identification and heart-mind development. When value and heart-mind signals are strong and clear, even if a task’s interest match is low, the recommendation system will still prioritize recommending it, merely adjusting the presentation form to reduce initial acceptance friction, and will never abandon a

task crucial to a student's heart-mind development simply because the student's interest does not lie in that particular theme.

3. Heart-Mind Bandit

The core decision problem of a recommendation system can be formulated as selecting the currently optimal item from a set of candidate contents according to a certain value criterion. Mathematically, this value criterion is formalized as a utility function, which serves as the ultimate basis for all decisions of the recommendation algorithm. The utility functions of traditional recommendation engines almost invariably consist only of a weighted sum of cognitive fit and interest match, allowing them to distinguish only the “completability” and “acceptability” of a task, but rendering them unable to judge its educational significance [4]. When a team collaboration task that can precisely target a student's compassion weakness but whose theme is not on her interest list competes with an easy reading task on a military topic, the traditional utility function will give the latter a higher score on the interest item, thereby easily eliminating the former. This is precisely the fundamental reason why this paper reconstructs the utility function.

The reconstructed heart-mind prioritized utility function consists of four weighted components and follows a strict priority order. The component with the highest weight is the value activation term, measuring whether a task can effectively awaken and deepen the student's specific Red Soul identification. For example, after a reverence experience triggered at a heart-mind anchor point in a VR red building experience, the system should preferentially recommend a study task that can continue and deepen their loyalty or responsibility soul, rather than switching to a creative activity unrelated to that value direction. The component with the second highest weight is the heart-mind targeting term, measuring whether the task precisely challenges the weaker dimensions within the student's current Nine-Force zone of proximal development—not simply recommending “resilience training,” but recommending a specific challenge that happens to be near the upper limit of the student's current resilience threshold and requires genuine effort to complete [7]. The component with the third weight is the cognitive fit term, which acts as a constraint to ensure that the cognitive complexity of the task does not exceed the student's processing capacity, preventing the cognitive threshold from blocking the student before the heart-mind quality is truly activated. The component with the lowest weight is the interest match term, serving as an auxiliary reference to help the

system select a more easily acceptable entrance form among multiple tasks with similar heart-mind benefits.

The design of this utility function completely overturns the “user first” logic that recommendation systems have followed since their inception at the algorithmic level. The traditional logic treats user clicks and dwell time as the final judge, with all the algorithm’s intelligence devoted to more accurately predicting these behavioral signals. The heart-mind prioritized logic, by contrast, declares that students’ subjective preferences are merely a reference; what truly determines recommendation priority is whether the content can most effectively promote the student’s heart-mind growth. This is the direct mathematical mapping of the educational stance that “soul-casting” takes precedence over “catering.”

The utility function establishes the direction of “what to recommend,” but does not yet answer the strategic question of “when and how to recommend.” The contextual bandit algorithm provides a classical mathematical tool for this problem by dynamically balancing “exploitation” of known high-quality options and “exploration” of options with higher uncertainty in each recommendation round to maximize cumulative long-term reward [8]. However, the reward signal in traditional bandit algorithms is immediate feedback of user behavior—clicks, purchases, dwell time. If the heart-mind recommendation system continues to use such signals, it will repeat the errors of commercial recommendation. Therefore, the core transformation of the bandit algorithm in this paper is the thorough replacement of the behavioral reward signal with heart-mind cultivation gain.

Heart-mind cultivation gain is not a value that can be directly chosen by the student, but an educational effectiveness metric automatically calculated by the system through multimodal perception data during and after task execution. Taking resilience as an example, the system recommends a Long March sandbox simulation task containing a resource-loss setback event. After task completion, the algorithm does not concern itself with whether the student clicked or replayed the task, but continuously analyzes her heart rate variability data, precisely calculating the half-life recovery time for the RMSSD indicator to return from its lowest point after the setback to 50% of the baseline level. This recovery time is compared with the student’s historical mean for the same indicator, and the amount of improvement in recovery speed—for instance, from an average of thirty-five seconds to twenty-two seconds—is mapped to the resilience cultivation gain for this recommendation. Similarly, concentration gain is synthesized from the magnitude of decrease in postural entropy and improvement in eye movement gaze stability during task execution; discernment gain is quantified by the

incidence of self-correction behaviors in codebook review tasks; and compassion gain is characterized by the frequency of anonymous peer nominations for helping behaviors after team collaboration. This physiological reconstruction of the reward signal makes the maximization objective pursued by the bandit algorithm precisely aligned with the fundamental goal of education in mathematical form.

Exploration and exploitation in the Heart-Mind Bandit are endowed with deeper educational meaning beyond merely reducing uncertainty. When a certain Nine-Force dimension in the student's heart-mind development vector has a confidence status of "to be confirmed" or "data blank," the system does not passively wait for data to accumulate naturally but actively triggers heart-mind exploration. This exploration is not a random task chosen to probe but deliberately prioritizes the task that can most effectively reveal the true level of that force. For a student with a persistent data blank in resilience, the system may push a moderately difficult sandbox setback and closely observe her behavioral and physiological responses. The challenge intensity of the exploration task is strictly limited within a safety boundary of "sufficient to expose the true level but not enough to cause severe frustration," and a teacher-guided review and reflection session is mandatorily embedded after task completion to ensure that even if the exploration result does not meet expectations, the student can still gain training at the level of discernment. The exploitation phase gradually transitions from consolidating strengths to compensating for weaknesses based on profile confidence. In the early stages of instruction, the system primarily exploits the high-identification dimensions already clear in the value vector, pushing positive challenges the student can meet, to build the self-efficacy needed for more arduous tasks; as the confidence of the heart-mind vector gradually increases, the focus of the exploitation strategy shifts to those qualities clearly in the to-be-developed range with high confidence, perfectly aligning with Vygotsky's zone of proximal development theory [7].

Moreover, the Heart-Mind Bandit possesses a critical capability that traditional recommendation systems lack, namely "active pausing." When the multimodal perception system continuously detects indicators such as a sustained decline in the student's HRV recovery rate, a steady rise in the skin conductance baseline, and an increase in avoidance postures reflected in postural entropy, these signals collectively point to the judgment that the student's resilience is approaching the exhaustion threshold. At this time, the algorithm does not continue to push more high-challenge tasks but actively pauses such recommendations and instead pushes recovery-oriented content such as concentration maintenance exercises,

compassion micro-experiences, or creativity free creation tasks. This decision to “realize that now it is appropriate to recommend less” is itself a high-level heart-mind protection strategy.

4. Monte Carlo Tree Search Path Planning under Narrative Constraints

The recommendation given by the Heart-Mind Bandit in each interaction round is a single-step optimum; it selects the task with the highest current utility based on the student’s latest four-dimensional profile. However, the historical knowledge graph that red education relies on possesses a tight internal structure: events have irreversible temporal sequences, causal chains are interlocking, and geographic spatial transfers must conform to historical logic. If the recommendation system only pursues heart-mind gain at each step, it might push the tactical analysis of the Zunyi Conference, which presupposes the Gutian Conference, before the student has understood the background of the Gutian Conference. At that moment, the algorithm may indeed achieve a good cultivation gain in adaptability, but the historical cognition constructed by the student suffers irreversible damage due to the rupture of the causal chain. This means that single-step optimality does not equate to full-path optimality; the recommendation system must possess a global perspective.

To obtain this global perspective, this paper introduces the Monte Carlo Tree Search (MCTS) algorithm for path planning on the knowledge graph [9]. The strength of MCTS lies in its ability to simultaneously balance immediate and long-term rewards through extensive simulation rollouts in a vast candidate path space, thereby selecting, at each actual decision step, the path that may not have the highest immediate return but is most advantageous in terms of long-term cumulative heart-mind benefits. We treat each node in the knowledge graph as a state in the search tree and each directed edge as a legal action. Each search iteration comprises the four standard phases of selection, expansion, simulation, and backpropagation, with each phase adapted to the specific needs of the educational scenario.

During the selection phase, the algorithm starts from the root node and recursively selects child nodes downward along the tree until it reaches a node that has not yet been fully expanded. The selection strategy employs an upper confidence bound formula, which consists of the sum of two parts: one is the average cumulative heart-mind gain obtained by that node in historical simulations, and the other is an exploration bonus term inversely proportional to the node’s visit count. This design ensures that the algorithm dynamically balances “taking the old path of known high returns” and “trying a new path that may lead to a completely new

direction of heart-mind cultivation.” In the expansion phase, the algorithm adds new candidate nodes according to the outgoing edges of the knowledge graph. Unlike the random expansion of standard MCTS, expansion here is guided by the semantic structure of the graph — priority is given to unfolding subsequent nodes marked with high heart-mind cultivation potential on the “Soul–Force–Quotient” transmission edges, while nodes whose cognitive difficulty far exceeds the student’s current resilience upper limit are pruned in advance. The simulation phase starts from the newly expanded node and employs a heuristic fast strategy to roll out along possible subsequent path sequences until a preset termination condition is met, such as the path length reaching the upper limit, the heart-mind benefit accumulation curve leveling off, or critical value anchor points having been covered. The backpropagation phase then propagates the cumulative heart-mind gain obtained from this simulation back upward along the path, updating the visit counts and value estimates of the nodes traversed, so that the choices of upper-level nodes are continuously corrected in subsequent iterations.

After thousands of iterations, the algorithm selects the direct child node of the root node with the highest visit count as the task actually recommended for the current step. Behind this decision, the global heart-mind cultivation benefits of the subsequent multiple steps have already been implicitly considered, rather than merely the immediate utility of the next step.

However, pure utility-maximization-driven search still carries a risk: MCTS might recommend a path with extremely high heart-mind cultivation benefits but that is logically untenable in terms of historical narrative. To prevent this risk, we formalize the core constraints of historical narrative as hard preconditions during the search process. Specifically, temporal consistency requires that the chronological order of any two adjacent event nodes on a path must conform to historical facts; jumps that reverse the flow of time are prohibited. For narratives that need to be presented in a thematic flashback manner, clear markers must be made on their respective timelines so that students can perceive the narrative’s retrospective movement during the experience. Causal consistency requires that, for two events already marked with a causal edge in the knowledge graph, no necessary intermediate links may be skipped; for events without annotated causal edges, causal chains must not be arbitrarily imposed and should instead be honestly marked as “temporal succession.” Spatial consistency requires that within the same temporal cross-section, the recommended path should not require the student to complete unreasonable large-span geographic transfers that exceed a reasonable threshold, unless the transfer has an explicit transfer edge annotation (such as a

Long March route annotation) in the knowledge graph. Any path violating one of the above constraints will have its cumulative reward value zeroed, meaning that branch will naturally be eliminated by the algorithm in subsequent iterations. By placing these constraints as filters in the pre-expansion phase, the seriousness of narrative logic becomes an absolute and insurmountable bottom line for heart-mind learning.

The path planning executes a continuous loop of “execute one step, re-perceive, re-search path.” After the student completes each task step, her behavioral performance and physiological data lead to updates of the heart-mind development vector and cognitive state vector. If the cognitive load indicator exceeds the warning threshold after task execution, the system will automatically lower the evaluation weight of the cognitive complexity dimension for subsequent candidate tasks in the new round of search. More critically, the path planned by MCTS is always defined as a “recommended path” rather than a “mandatory path” when presented to the student. The student retains full freedom of choice at every narrative branch point; the system merely provides a highlighted prompt for the recommended next node. When the student actively chooses to deviate from the recommended path, the system does not treat it as an error, imposes no penalty, and does not forcibly correct it, but takes the student’s chosen node as the new root and immediately re-plans in the remaining knowledge graph space. Personalization and autonomy thus coexist: the AI silently calculates the optimal sequence of heart-mind growth in the background, while the student herself always takes each step.

5. Experimental Analysis

To verify the effectiveness of the proposed framework, a prototype system was deployed in an OMOEC course at a red education practice base, with a total of 128 students participating in the experiment. Baseline methods included item-based collaborative filtering, linear contextual bandit, and MCTS path planning using only the cognition–interest utility function. Evaluation metrics included the standardized change values of the Nine-Force scale before and after the course, cumulative sum of heart-mind cultivation gains, historical causal consistency ratio of paths, as well as auxiliary indicators such as task persistence completion rate and interest match degree.

Experimental results showed that students in the experimental group achieved effect sizes of 0.71, 0.65, and 0.58 for resilience, concentration, and discernment respectively, all significantly superior to the linear bandit baseline. This indicates that using heart-mind

cultivation gain quantified by physiological indicators as the reward signal can indeed guide the algorithm to continuously recommend challenging tasks that truly stretch students' heart-mind weaknesses. Notably, the interest match degree of the experimental group decreased slightly by about eight percentage points compared to collaborative filtering, yet students' task persistence completion rate increased by twelve percentage points. This seemingly contradictory phenomenon reveals an important educational principle: when the recommended content precisely touches upon the student's internal growth needs, even if it is slightly inferior in entertainment appeal, it can still inspire a more resilient motivation to persist.

Regarding narrative logic assurance, the MCTS path planning incorporating the three hard constraints achieved 100% historical causal consistency in its recommended paths. In contrast, the control MCTS driven purely by utility without hard constraints showed temporal disorder or causal link skipping in approximately 23% of paths. In post-class student interviews, multiple students reported that learning along the recommended path “felt like experiencing the real history in person,” and the light narrative interludes interspersed between event nodes—such as breath anchoring or micro-reviews—did not interrupt the rhythm but instead enhanced their internal solemnity before entering the next major historical event. These qualitative feedbacks together with the quantitative data demonstrate that heart-mind priority and narrative constraints can be effectively balanced within a unified algorithmic framework.

6. Conclusion

This paper constructs a decision framework that comprehensively shifts AI recommendation from “interest catering” to “heart-mind cultivation.” The four-dimensional student profile formally incorporates values and heart-mind qualities into the perception space of the recommendation system, the Heart-Mind Bandit replaces the traditional behavioral reward signal with Nine-Force cultivation gain quantified by physiological and behavioral data, Monte Carlo Tree Search achieves global optimization of long-term educational paths under the protection of temporal, causal, and spatial hard constraints, and the autonomy preserved throughout ensures that human agency is not usurped by the algorithm. This framework not only provides an interpretable and auditable algorithmic foundation for holistic personality education but also fundamentally reshapes the role of AI in education: it no longer predicts and caters to desires but silently calculates every possible path toward a more complete personality and solemnly returns the right of choice to the learner at every fork. Future work

will aim to introduce blockchain notarization technology so that every recommendation decision and its basis can be immutably recorded and traced, while also exploring open-ended narrative generation based on large language models to further enrich the forms and depth of heart-mind cultivation tasks.

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